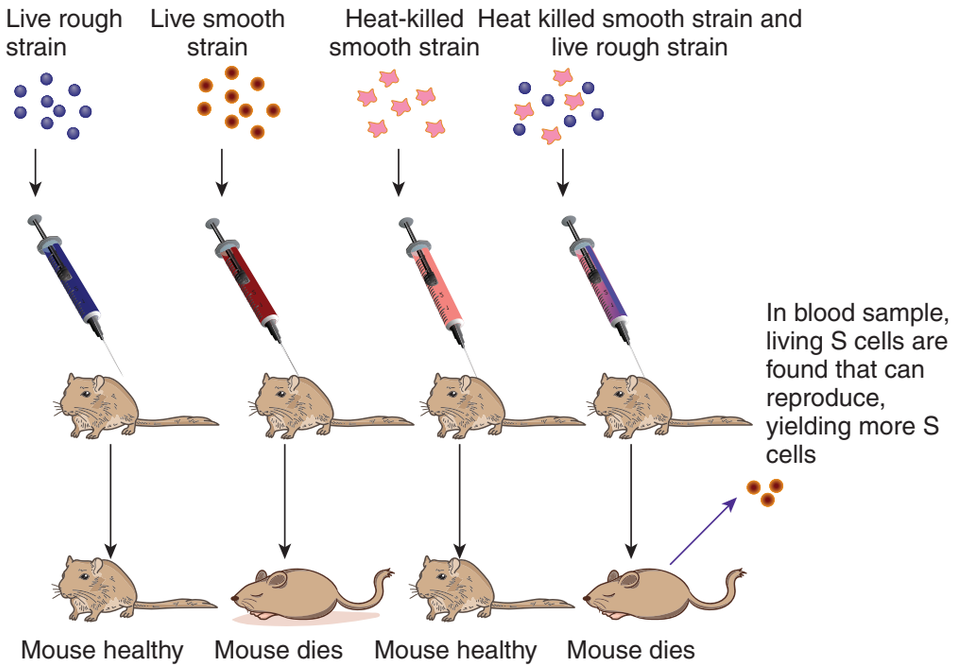


Figure 14.3: Summary of Griffith's experiment

HOTS

What did Griffith expect to happen to mouse when he injected it with live rough strain and heat killed smooth strain

14.1.2 Oswald T. Avery, Colin Mac Leod and Maclyn Mc Carty's Experiment



Figure 14.4: Avery *et al.,*

Griffith's experimental results led to the curiosity to explore the transforming principle. Avery and his colleagues (Figure 14.4) used the extracts of heat-killed smooth bacteria and treated it with enzyme protease, RNase, DNase to eliminate proteins, RNA and DNA respectively. Each of the treated extracts were mixed with live rough bacteria and injected into mice. The mice injected with a mixture of DNase treated extract and live rough strain did not die. This partially proved that DNA was responsible for changing the rough strain of *Streptococcus pneumoniae* bacteria into smooth bacteria. Avery *et al.,* experiment is summarized in Figure 14.5. Later Hershey and Chase's experiment on T2 bacteriophage confirmed that genetic information is present in DNA.

These important early experiments and many other lines of evidence have shown that DNA bears the genetic information of living cells and it is responsible for transfer of characteristics from one generation to another. This is true in all organisms, the notable exceptions being RNA viruses

TN_GOVT_XI_Micro_Biology_CH14.indd 255 16-02-2019 10.37.50 AM